

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P1H Worked Solutions

Jan 2020 | P1H

25 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

- 1** The point A has coordinates $(5, -4)$
The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB .

(.....,)
(2)

Line L has equation $y = 2 - 3x$

(b) Write down the gradient of line L .

.....
(1)

Line L has equation $y = 2 - 3x$

- (c) Does the point with coordinates $(100, -302)$ lie on line L ?
You must give a reason for your answer.

.....
(1)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 4

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Jan 2020 P1H - Jan 2020 | P1H

- 2** Find the lowest common multiple (LCM) of 28 and 105

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Area & Perimeter**

Jan 2020 P1H - Jan 2020 | P1H

- 3 The diagram shows a shape.

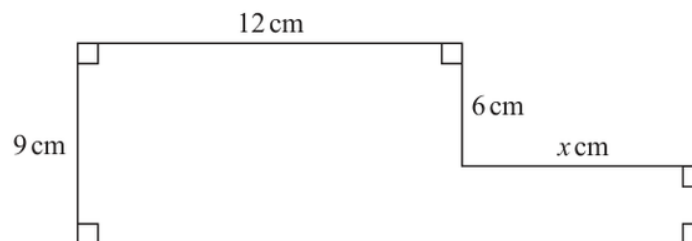


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

$x =$

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 4

TOPIC: **Area & Perimeter**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4** **Marks: 7****TOPIC: Statistics Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean weight of the 40 babies.

..... kg
(4)

One of the 40 babies is going to be chosen at random.

- (c) Find the probability that this baby has a weight of more than 5 kg.

.....
(2)

(Total for Question 4 is 7 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 7

TOPIC: **Statistics Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 6****TOPIC: Ratio Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

- 5** 120 children go on an activity holiday.
The ratio of the number of girls to the number of boys is 3 : 5
- On Sunday, all the children either go sailing or go climbing.
- $\frac{16}{25}$ of the boys go climbing.
- Twice as many girls go sailing as go climbing.
- Work out how many children go sailing on Sunday.

(Total for Question 5 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 5

Marks: 6

TOPIC: **Ratio Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2020 P1H - Jan 2020 | P1H

- 6 (a) Write 7.8×10^{-4} as an ordinary number.

.....
(1)

- (b) Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

.....
(2)

(Total for Question 6 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 9****TOPIC: Solving Linear Equations**

Jan 2020 P1H - Jan 2020 | P1H

7 (a) Expand and simplify $(m - 8)(m + 5)$
(2)(b) Factorise fully $5y + 20y^2$
(2)(c) Simplify $(p^2 + 3)^0$
(1)(d) Solve $3(2x - 5) = \frac{9 - x}{2}$

Show clear algebraic working.

 $x =$
(4)**(Total for Question 7 is 9 marks)**

PAST PAPER SOLUTION

Q#: 7

Marks: 9

TOPIC: **Solving Linear Equations**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

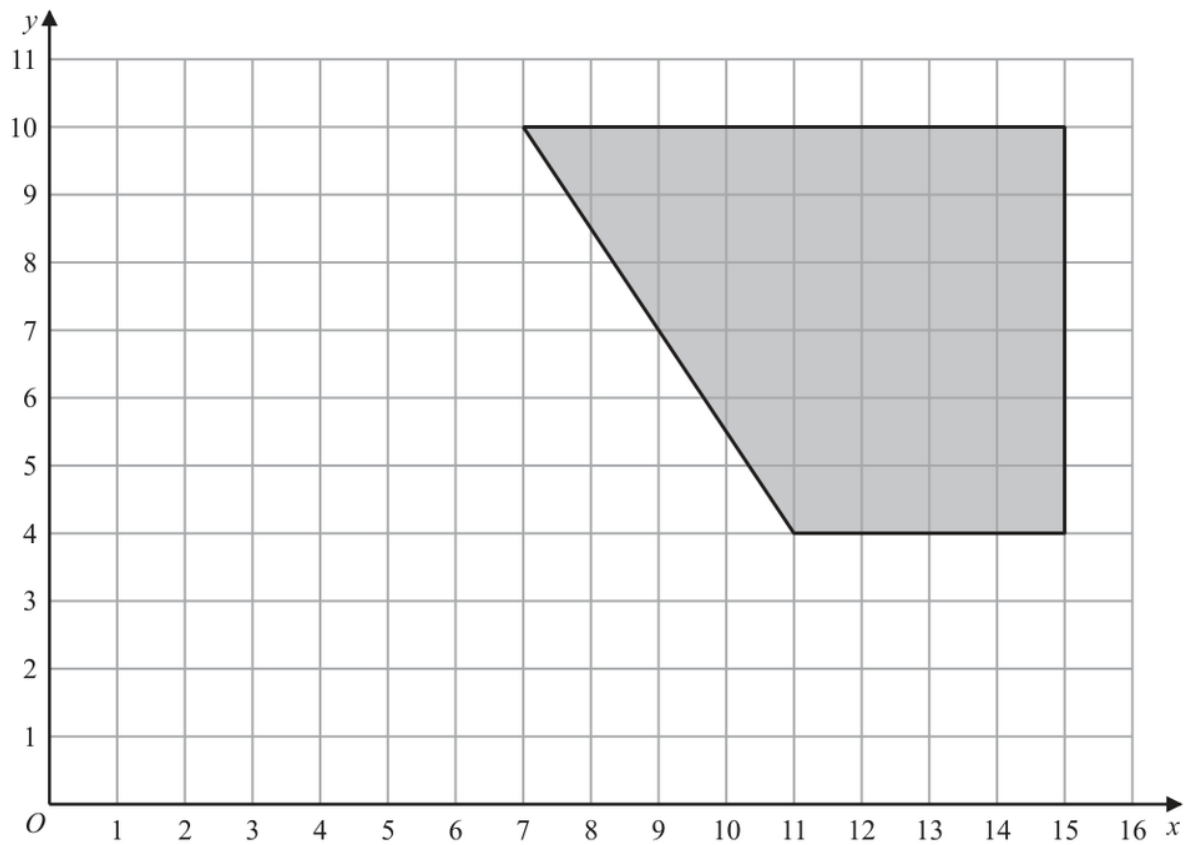
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 2****TOPIC: Transformations**

Jan 2020 P1H - Jan 2020 | P1H

8

On the grid, enlarge the shaded shape with scale factor $\frac{1}{2}$ and centre (1, 2)

(Total for Question 8 is 2 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 2

TOPIC: **Transformations**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

9 Here is a right-angled triangle.

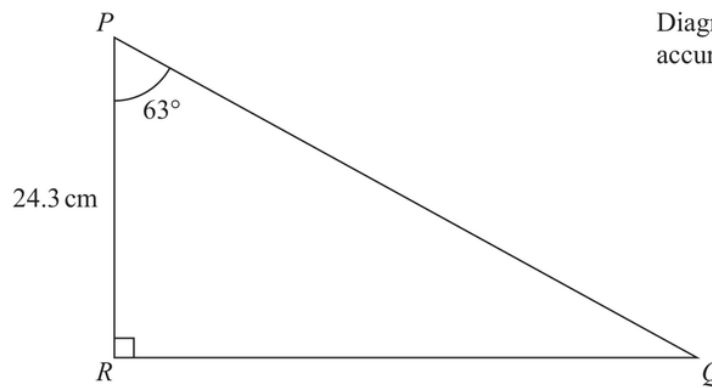


Diagram **NOT**
accurately drawn

Calculate the length of PQ .
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

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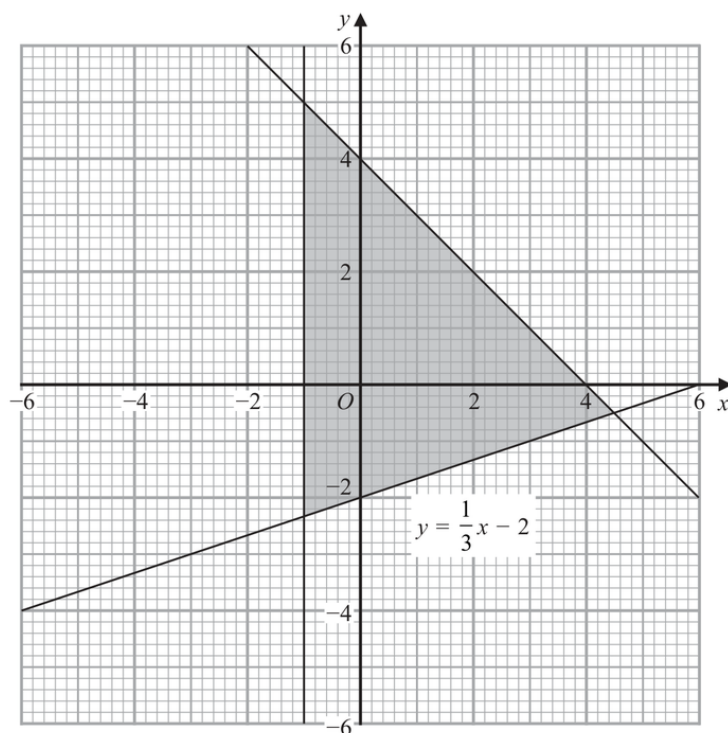
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10** **Marks: 3****TOPIC: Graphing Inequalities**

Jan 2020 P1H - Jan 2020 | P1H

- 10** The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

.....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Graphing Inequalities**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2020 P1H - Jan 2020 | P1H

- 11** Max invests \$6000 in a savings account for 3 years.
The account pays compound interest at a rate of 1.5% per year for the first 2 years.
- The compound interest rate changes for the third year.
At the end of 3 years, there is a total of \$6311.16 in the account.
- Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

..... %

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jan 2020 P1H - Jan 2020 | P1H

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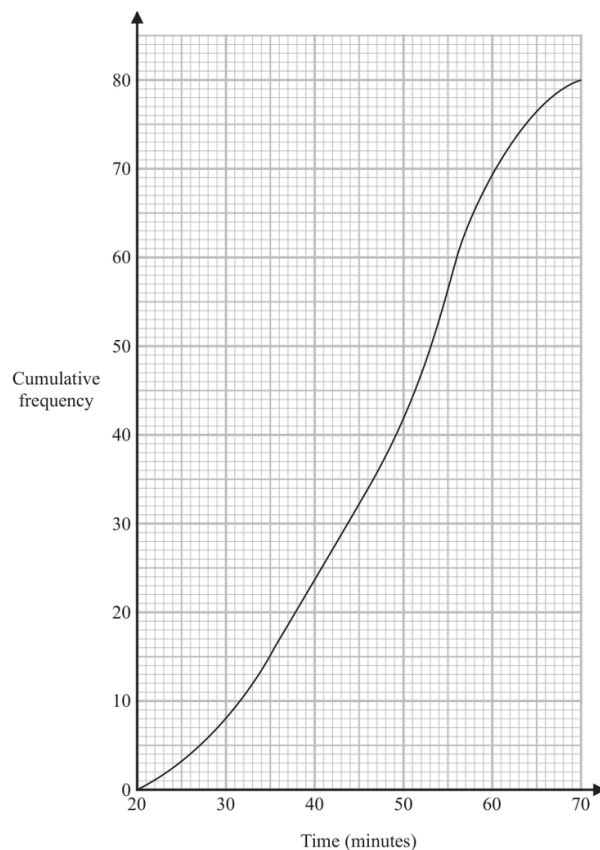
No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2020 P1H - Jan 2020 | P1H

- 12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



- (a) Use the graph to find an estimate for the interquartile range.

..... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

- (b) Work out an estimate for the number of men who took part in the race.

.....
(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Standard & Compound Units**

Jan 2020 P1H - Jan 2020 | P1H

13 The diagram shows a solid cube.

The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

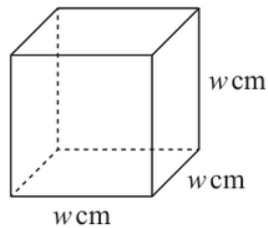


Diagram **NOT**
accurately drawn

The cube exerts a force of 56 newtons on the table.

The pressure on the table due to the cube is 0.14 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

..... cm³

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 4

TOPIC: **Standard & Compound Units**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

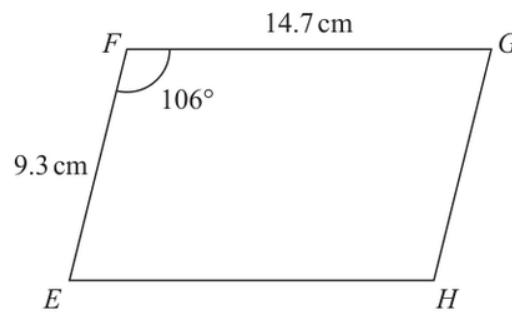
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1H - Jan 2020 | P1H

14 The diagram shows parallelogram $EFGH$.Diagram **NOT**
accurately drawn

$EF = 9.3 \text{ cm}$

$FG = 14.7 \text{ cm}$

$\text{Angle } EFG = 106^\circ$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

- (b) Work out the length of the diagonal EG of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION

Q#: 14

Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

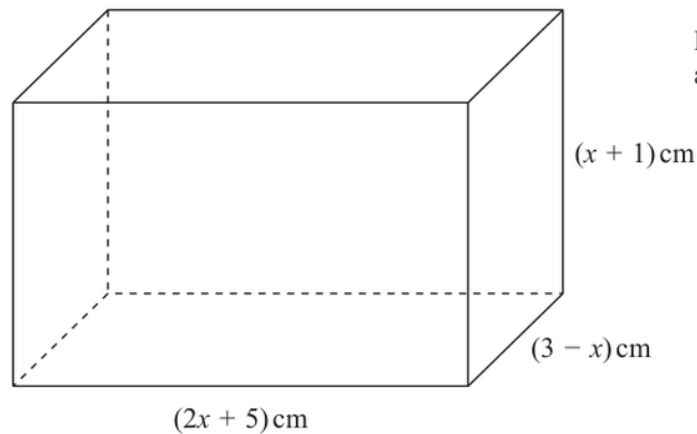
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 8****TOPIC: Differentiation**

Jan 2020 P1H - Jan 2020 | P1H

15Diagram **NOT**
accurately drawnThe diagram shows a cuboid of volume $V \text{ cm}^3$ (a) Show that $V = 15 + 16x - x^2 - 2x^3$

(3)

There is a value of x for which the volume of the cuboid is a maximum.

- (b) Find this value of x .
 Show your working clearly.
 Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$
 (5)
(Total for Question 15 is 8 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 8

TOPIC: **Differentiation**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2020 P1H - Jan 2020 | P1H

16 $P = \frac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.

$c = 20$ correct to 2 significant figures.

$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P .

Show your working clearly.

Give your answer correct to 2 decimal places.

.....
(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 6****TOPIC: Algebraic Roots & Indices**

Jan 2020 P1H - Jan 2020 | P1H

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

(3)

(Total for Question 17 is 6 marks)

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PAST PAPER SOLUTION

Q#: 17

Marks: 6

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Probability Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

18 There are 16 sweets in a bowl.

4 of the sweets are blackcurrant.

5 of the sweets are lemon.

7 of the sweets are orange.

Anna, Ravi and Sam each take at random one sweet from the bowl.

Work out the probability that the 5 lemon sweets are still in the bowl.

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **Probability Toolkit**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

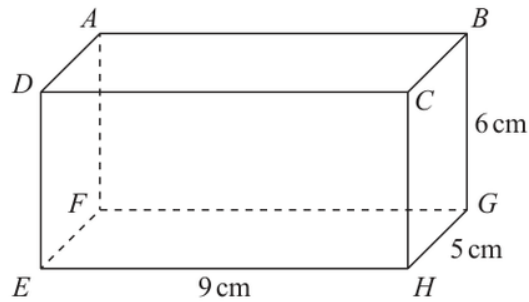
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

19 The diagram shows a cuboid $ABCDEFGH$.Diagram **NOT**
accurately drawn $EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19

Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Graphs of Functions**

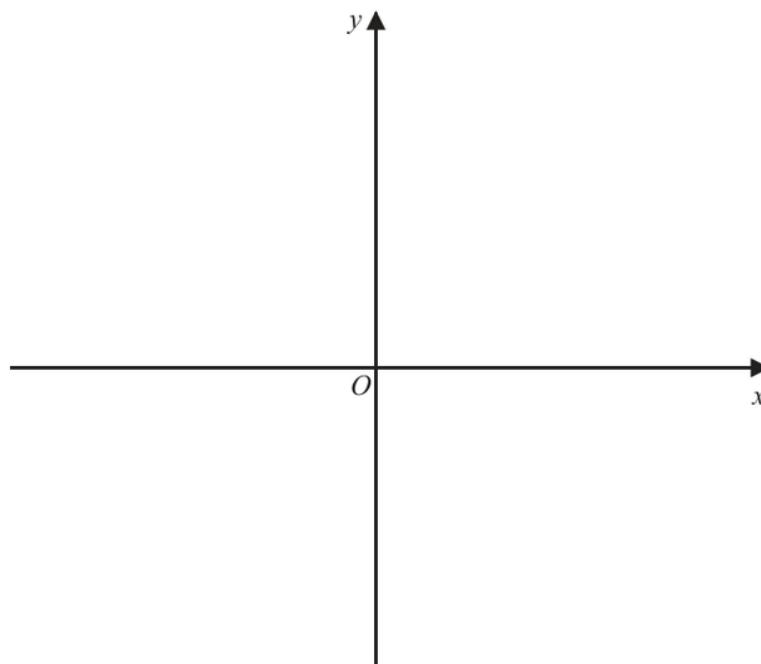
Jan 2020 P1H - Jan 2020 | P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.

On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Graphs of Functions**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Graphs of Functions**

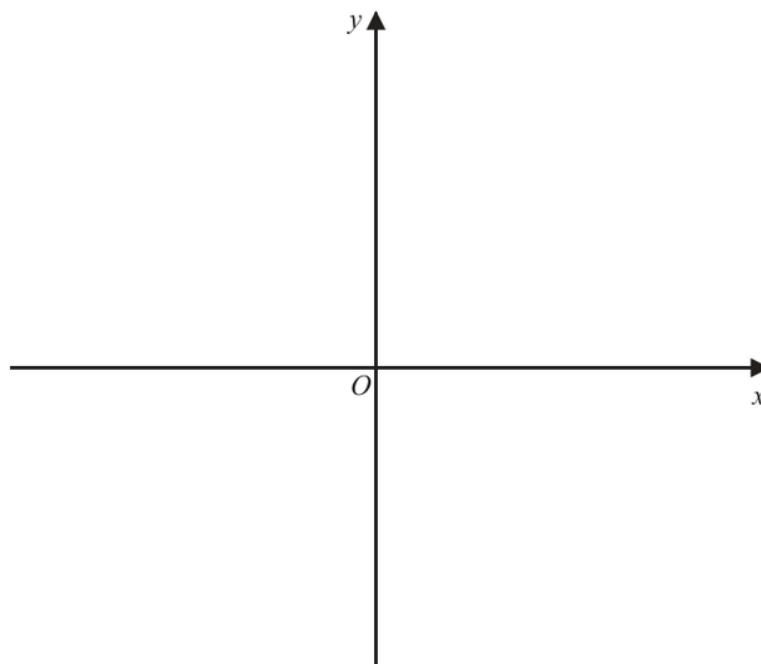
Jan 2020 P1H - Jan 2020 | P1H

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Using the axes below, sketch the curve **C**.

On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Graphs of Functions**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Functions**

Jan 2020 P1H - Jan 2020 | P1H

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Functions**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Functions**

Jan 2020 P1H - Jan 2020 | P1H

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Functions**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **Coordinate Geometry**

Jan 2020 P1H - Jan 2020 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA